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Three-dimensional simplified and unconditionally stable lattice Boltzmann method for incompressible isothermal and thermal flows

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In this paper, a three-dimensional simplified and unconditionally stable lattice Boltzmann method (3D-USLBM) is proposed for simulating incompressible isothermal/thermal flows. This method is developed by reconstructing solutions to the macroscopic governing equations recovered from the lattice Boltzmann equation and resolved in a predictor-corrector scheme. The final formulations of 3D-USLBM only involve the equilibrium and the non-equilibrium distribution functions. Among them, the former is calculated from the macroscopic variables and the latter is evaluated from the difference between two equilibrium distribution functions at different locations and time levels. Thus, 3D-USLBM directly tracks the evolution of macroscopic variables, which yields lower cost in virtual memory and facilitates the implementation of physical boundary conditions. A von Neumann stability analysis was performed on the present method to theoretically prove its unconditional stability. By imposing a regular Lagrange interpolation algorithm, this method can be flexibly extended to a non-uniform Cartesian mesh or body-fitted mesh with curved boundaries. Four numerical tests, that is, plane Poiseuille flow, 3D lid-driven cavity flow and 3D natural convection in a cubic cavity, and concentric annulus, were conducted to verify the stability, accuracy, and flexibility of the presented method. *Published by AIP Publishing.* [http://dx.doi.org/10.1063/1.4983339]

I. INTRODUCTION

Incompressible isothermal/thermal flow has always been an important topic in fluid mechanics.[1–3] Computational Fluid Dynamics (CFD) has been proven to be an effective approach to solve it.[4] Within the CFD framework, the lattice Boltzmann method (LBM) is an emerging method that has attracted increasing attention.[5–25] Basically, LBM is a mesoscopic method originating from the lattice Boltzmann equation (LBE). Unlike the general approach of macroscopic methods which performs conservation laws on the control volume, LBM introduces the distribution functions to describe the motion and the distribution of the fluid particles. Interactions among fluid particles are realized through the so-called streaming-collision process which can be approximated by a simple relationship given by the Bhatnagar-Gross-Krook (BGK) model.[26] Simplicity and explicitness are two main advantages of LBM. Specifically, LBM describes the complex fluid phenomenon by very simple algebraic equations, and the pressure can be explicitly obtained through the equation of state (EOS) without introducing a tedious pressure Poisson equation in macroscopic solvers.

However, it is also noted that the standard LBM with the second order of accuracy suffers from several drawbacks, some of which may be exacerbated due to implementation strategies rather than the numerical model itself. First, it requires large amounts of virtual memory, especially in three-dimensional simulations.[27] Theoretically, distribution functions along all lattice velocity directions at each mesh point need to be stored. For a typical D3Q15 model, at least 15 variables are required for each mesh point in isothermal simulations. Second, physical boundary conditions need to be transformed into conditions of the distribution functions to be implemented in the standard LBM.[28–31] Such transformations require extra effort when encountering problems with curved boundaries. Another drawback of the standard second-order LBM is its poor numerical stability at high Reynolds numbers or high Rayleigh numbers,[32–36] in which very fine meshes are usually adopted to ensure the convergence of the computation. And finally, because of the lattice uniformity introduced by the symmetric lattice velocity model, the standard LBM can only be directly implemented on uniform meshes. To extend it to non-uniform meshes, extra interpolation/fitting algorithms are inevitable.[13,15,37–39]

To eliminate or alleviate some of the above drawbacks of the standard lattice Boltzmann method (LBM), a simplified lattice Boltzmann method (SLBM) was recently developed.[40,41] However, detailed derivations show that the SLBM can only ensure the first-order accuracy in time marching, and the work on extension to three-dimensions remains unfinished.[40] To improve time-marching accuracy in SLBM while maintaining its merits, in this paper, we present a three-dimensional simplified and unconditionally stable lattice Boltzmann method (3D-USLBM). The derivation of 3D-USLBM is firmly based on the classic Chapman-Enskog (C-E) expansion analysis which works as the link between the lattice Boltzmann equation (LBE) and macroscopic Navier-Stokes (N-S) equations.[42,43] Through C-E analysis, it can be proven that LBE can recover macroscopic equations to the second-order of accuracy both in

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space and in time. In the present work, the recovered macroscopic equations are resolved by a predictor-corrector scheme and their solutions are reconstructed within the LBM framework by using lattice properties and conservation laws. The final formulation of 3D-USLBM involves only the equilibrium and the non-equilibrium distribution functions. Specifically, the equilibrium distribution function can be calculated from macroscopic variables, while the non-equilibrium part is approximated by two equilibrium distribution functions at different locations and time levels. Such treatment of the non-equilibrium term is inspired by a recent work of Wang *et al.*[44–46] on a lattice Boltzmann flux solver (LBFS) and helps to avoid tedious expansion terms from previous work. By doing so, the present 3D-USLBM directly tracks the evolution of macroscopic variables rather than the distribution functions, which greatly saves virtual memory and facilitates the implementation of physical boundary conditions. Additionally, through von Neumann stability analysis, the present method can be proven unconditionally stable. This endows it with significant advantages over the standard LBM, especially in problems at high Reynolds/Rayleigh numbers.

The remaining parts of this paper are laid out as follows: Section II gives a brief introduction to the Chapman Enskog (C-E) expansion analysis that connects the lattice Boltzmann equation and the macroscopic governing equations, which is the theoretical foundation of the present work. A detailed derivation of the three-dimensional simplified and unconditionally stable lattice Boltzmann method (3D-USLBM), together with its boundary treatments and von Neumann stability analysis, is offered in Section III. In Section IV, four representative tests of incompressible isothermal/thermal flows are carried out by 3D-USLBM to validate its accuracy, stability, and flexibility on non-uniform meshes. Conclusions are finally drawn in Section IV.

II. THEORETICAL FOUNDATIONS

The theoretical foundation of the method proposed in this paper is the Chapman-Enskog (C-E) expansion analysis that connects two systems of mathematical interpretation of fluid problems, i.e., lattice Boltzmann equation (LBE) and macroscopic Navier-Stokes (N-S) equations. In this section, these two systems are briefly introduced. Lattice properties and relationships given by the C-E analysis that will be used in later derivations of 3D-USLBM are also presented.

A. Macroscopic governing equations

For general thermal fluid problems, by imposing conservation laws on the fluid control volume, one can derive the following macroscopic governing equations:[1]

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \rho \mathbf{u} = 0, \tag{1}$$

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot (\rho \mathbf{u}\mathbf{u}) = -\nabla p + \upsilon \nabla \cdot \left[\nabla \rho \mathbf{u} + (\nabla \rho \mathbf{u})^T\right] + \mathbf{F}_E, \tag{2}$$

$$\frac{\partial}{\partial t}(\rho e) + \nabla \cdot (\rho \mathbf{u} e) = \chi \nabla^2 (\rho e), \tag{3}$$

where ρ, $\mathbf{u}$, p, and υ denote the density, the velocity vector, the pressure, and the kinematic viscosity, respectively. $\mathbf{F}_E$ and χ are the external force term and the thermal diffusivity, respectively, and e is the internal energy defined as $e = DRT/2$. D, R, and T are the number of spatial dimensions, the gas constant, and the temperature. Here, we set $D = 3$ and $R = 1$ in reference to previous studies.[27,45]

For an isothermal viscous problem, without considering thermal effects, the flow is simulated simply by solving Eqs. (1) and (2), and $\mathbf{F}_E$ is set to zero if no external force is considered. When considering thermal effects, the energy equation (3) is incorporated into the system of equations to be solved. In such case, the velocity field has a direct impact on the energy, since it is involved in the convection term of the energy equation. The thermal effect on the velocity field can be reflected in an external force term in the momentum equation by using the Boussinesq approximation,[16,47]

$$\mathbf{F}_E = \begin{pmatrix} 0 \\ 0 \\ -\rho g \beta (T - T_m) \end{pmatrix}, \tag{4}$$

where g, β, and T_m are the gravitational acceleration, the thermal expansion coefficient, and the average temperature, respectively.

B. Thermal lattice Boltzmann method

Various thermal lattice Boltzmann (LB) models have been proposed and applied in simulations. One popular model is the thermal energy distribution model.[24,25,48] In this model, the internal energy distribution function is adopted to incorporate the thermal effects. The evolution of the distribution functions can be expressed as

$$f_\alpha(\mathbf{r} + \mathbf{e}_\alpha \delta_t,\ t + \delta_t) = f_\alpha(\mathbf{r},\ t) + \frac{f_\alpha^{eq}(\mathbf{r},\ t) - f_\alpha(\mathbf{r},\ t)}{\tau_\upsilon}, \quad \alpha = 0,\ 1,\ \ldots,\ M, \tag{5}$$

$$g_\alpha(\mathbf{r} + \mathbf{e}_\alpha \delta_t,\ t + \delta_t) = g_\alpha(\mathbf{r},\ t) + \frac{g_\alpha^{eq}(\mathbf{r},\ t) - g_\alpha(\mathbf{r},\ t)}{\tau_c}, \quad \alpha = 0,\ 1,\ \ldots,\ N, \tag{6}$$

where f_α and g_α are the density distribution function and the internal energy distribution function along the α direction, respectively. The superscript "eq" denotes the equilibrium state, τ_υ and τ_c are the single relaxation parameters related to the kinematic viscosity and the thermal diffusivity, respectively, δ_t is the time step applied in the model, and M and N are the number of the lattice velocities used in the density and the internal energy distribution functions, respectively. In thermal LBM, the macroscopic properties can be obtained from the conservation laws of mass, momentum, and energy:[5,6]

$$\rho = \sum_\alpha f_\alpha = \sum_\alpha f_\alpha^{eq}, \quad \rho \mathbf{u} = \sum_\alpha f_\alpha \mathbf{e}_\alpha = \sum_\alpha f_\alpha^{eq} \mathbf{e}_\alpha, \quad \rho e = \sum_\alpha g_\alpha = \sum_\alpha g_\alpha^{eq}. \tag{7}$$

The thermal lattice Boltzmann method is associated with the lattice velocity model. In the present work, a typical 15-bit lattice velocity model (D3Q15) is applied (see Figure 1),[7] which consists of the following lattice velocity directions:

$$\mathbf{e}_\alpha = \begin{cases} 0, & \alpha = 0 \\ (\pm 1,\ 0,\ 0),\ (0,\ \pm 1,\ 0),\ (0,\ 0,\ \pm 1), & \alpha = 1 \sim 6 \\ (\pm 1,\ \pm 1,\ \pm 1), & \alpha = 7 \sim 14 \end{cases}. \tag{8}$$

The equilibrium distribution functions are given by

$$f_\alpha^{eq} = \rho\omega_\alpha \left[1 + \frac{\mathbf{e}_\alpha \cdot \mathbf{u}}{c_s^2} + \frac{(\mathbf{e}_\alpha \cdot \mathbf{u})^2 - (c_s\,|\mathbf{u}|)^2}{2c_s^4}\right], \quad \alpha = 0, 1, \ \ldots, \ 14, \tag{9}$$

$$g_\alpha^{eq}(\mathbf{r}, t) = \begin{cases} -\dfrac{\rho e}{3}\dfrac{|\mathbf{u}|^2}{c^2}, & \alpha = 0 \\ \dfrac{\rho e}{9}\left[1 + \dfrac{\mathbf{e}_\alpha \cdot \mathbf{u}}{c^2} + \dfrac{9}{2}\dfrac{(\mathbf{e}_\alpha \cdot \mathbf{u})^2}{c^4} - \dfrac{3}{2}\dfrac{|\mathbf{u}|^2}{c^2}\right], & \alpha = 1 - 6 \\ \dfrac{\rho e}{72}\left[3 + 7 \cdot \dfrac{\mathbf{e}_\alpha \cdot \mathbf{u}}{c^2} + \dfrac{9}{2}\dfrac{(\mathbf{e}_\alpha \cdot \mathbf{u})^2}{c^4} - \dfrac{3}{2}\dfrac{|\mathbf{u}|^2}{c^2}\right], & \alpha = 7 - 14 \end{cases}, \tag{10}$$

where $c = \delta_x/\delta_t$ is set to be 1 in this work, and δ_x is the lattice spacing. The weighting coefficients ω_α and the sound speed c_s are

$$\omega_0 = \frac{2}{9}, \quad \omega_{1-6} = \frac{1}{9}, \quad \omega_{7-14} = \frac{1}{72}, \quad c_s = \frac{c}{\sqrt{3}}. \tag{11}$$

C. Chapman-Enskog (C-E) expansion analysis

Chapman-Enskog (C-E) expansion analysis connects the lattice Boltzmann equation to the macroscopic governing equations.[8,42] By employing the C-E analysis on the lattice Boltzmann equations (5) and (6), the following equations can be recovered:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \left(\sum_\alpha \mathbf{e}_\alpha f_\alpha^{eq}\right) = 0, \tag{12}$$

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot \Pi = 0, \tag{13}$$

$$\frac{\partial \rho e}{\partial t} + \nabla \cdot \mathbf{Q} = 0, \tag{14}$$

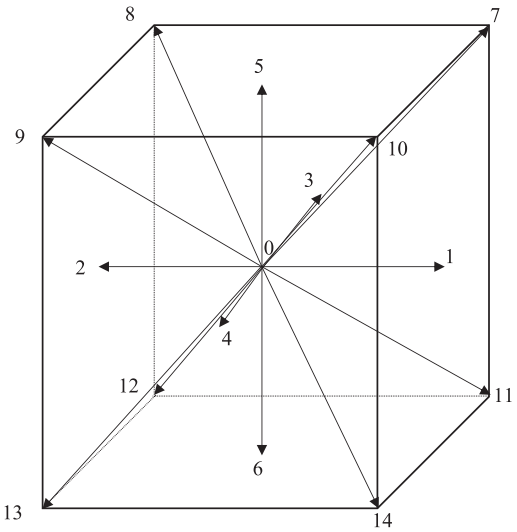


FIG. 1. Illustration of the D3Q15 model.

where Π and $\mathbf{Q}$ are the momentum flux tensor and the thermal flux vector in the following form:

$$\Pi_{\beta\gamma} = \sum_\alpha (\mathbf{e}_\alpha)_\beta (\mathbf{e}_\alpha)_\gamma \left[f_\alpha^{eq} + \left(1 - \frac{1}{2\tau_\upsilon}\right)\varepsilon f_\alpha^{(1)}\right], \tag{15}$$

$$\mathbf{Q} = \sum_\alpha \mathbf{e}_\alpha \left[g_\alpha^{eq} + \left(1 - \frac{1}{2\tau_c}\right)\varepsilon g_\alpha^{(1)}\right], \tag{16}$$

where $(\mathbf{e}_\alpha)_\beta$ and $(\mathbf{e}_\alpha)_\gamma$ are the components of the lattice velocity vector $\mathbf{e}_\alpha$ in the β-direction and in the γ-direction, respectively. The non-equilibrium terms $\varepsilon f_\alpha^{(1)}$ and $\varepsilon g_\alpha^{(1)}$ fulfill the following relationships:

$$f_\alpha^{neq} = f_\alpha - f_\alpha^{eq} \approx \varepsilon f_\alpha^{(1)} = -\tau_\upsilon \delta_t \left(\frac{\partial}{\partial t} + \mathbf{e}_\alpha \cdot \nabla\right) f_\alpha^{eq} = -\tau_\upsilon \delta_t \mathrm{D} f_\alpha^{eq}, \tag{17}$$

$$g_\alpha^{neq} = g_\alpha - g_\alpha^{eq} \approx \varepsilon g_\alpha^{(1)} = -\tau_c \delta_t \left(\frac{\partial}{\partial t} + \mathbf{e}_\alpha \cdot \nabla\right) g_\alpha^{eq} = -\tau_c \delta_t \mathrm{D} g_\alpha^{eq}, \tag{18}$$

where the superscript "*neq*" represents the non-equilibrium distribution function, and the operator D denotes $D = \left(\frac{\partial}{\partial t} + \mathbf{e}_\alpha \cdot \nabla\right)$. It can be proven that Eqs. (12)–(14) are equivalent to the macroscopic governing equations (1)–(3), given the following relationships between relaxation parameters, the kinematic viscosity and the thermal diffusivity:

$$\upsilon = c_s^2 \delta_t \left(\tau_\upsilon - \frac{1}{2}\right), \tag{19}$$

$$\chi = \frac{5}{3} c^2 \delta_t \left(\tau_c - \frac{1}{2}\right). \tag{20}$$

III. THREE-DIMENSIONAL UNCONDITIONALLY STABLE LATTICE BOLTZMANN METHOD (3D-USLBM)

The aforementioned drawbacks of the standard LBM, including high cost in virtual memory, inconvenient implementation of physical boundary conditions, and poor numerical

stability at a high Reynolds number or high Rayleigh number, constrain it from broader applications to some extent. Moreover, in three-dimensional scenarios, more lattice vector directions are adopted and more complex boundaries are encountered, which magnify these drawbacks. To tackle these problems, we present a three-dimensional simplified and unconditionally stable lattice Boltzmann method (3D-USLBM).

A. Formulations

By adopting the fractional-step technique,[49,50] the macroscopic governing equations (12)–(14) recovered from the lattice Boltzmann equation through the Chapman-Enskog (C-E) expansion analysis can be solved in the following predictor-corrector scheme.

Predictor step:

$$\frac{\partial \rho}{\partial t} + \nabla \cdot \left(\sum_{\alpha} \mathbf{e}_{\alpha} f_{\alpha}^{eq} \right) = 0, \tag{21}$$

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot \left[\sum_{\alpha} (\mathbf{e}_{\alpha})_{\beta} (\mathbf{e}_{\alpha})_{\gamma} f_{\alpha}^{eq} + \frac{1}{2\tau_{\upsilon}} \sum_{\alpha} (\mathbf{e}_{\alpha})_{\beta} (\mathbf{e}_{\alpha})_{\gamma} f_{\alpha}^{neq} \right] = 0, \tag{22}$$

$$\frac{\partial \rho e}{\partial t} + \nabla \cdot \left[\sum_{\alpha} \mathbf{e}_{\alpha} g_{\alpha}^{eq} + \frac{1}{2\tau_{c}} \sum_{\alpha} \mathbf{e}_{\alpha} g_{\alpha}^{neq} \right] = 0. \tag{23}$$

Corrector step:

$$\frac{\partial \rho}{\partial t} = 0, \tag{24}$$

$$\frac{\partial \rho \mathbf{u}}{\partial t} + \nabla \cdot \left[\left(1 - \frac{1}{\tau_{\upsilon}} \right) \sum_{\alpha} (\mathbf{e}_{\alpha})_{\beta} (\mathbf{e}_{\alpha})_{\gamma} f_{\alpha}^{neq} \right] - \mathbf{F}_{E} = 0, \tag{25}$$

$$\frac{\partial \rho e}{\partial t} + \nabla \cdot \left[\left(1 - \frac{1}{\tau_{c}} \right) \sum_{\alpha} \mathbf{e}_{\alpha} g_{\alpha}^{neq} \right] = 0. \tag{26}$$

By using lattice properties and relationships given by the C-E analysis, solutions to the above equations can be reconstructed within the LB framework. In particular, the solutions to the equations in the predictor step, namely, intermediate density, velocity and energy, are evaluated by

$$\rho^{*} = \sum_{\alpha} f_{\alpha}^{eq} (\mathbf{r} - \mathbf{e}_{\alpha} \delta_{t}, t - \delta_{t}), \tag{27}$$

$$\rho^{*} \mathbf{u}^{*} = \sum_{\alpha} \mathbf{e}_{\alpha} f_{\alpha}^{eq} (\mathbf{r} - \mathbf{e}_{\alpha} \delta_{t}, t - \delta_{t}), \tag{28}$$

$$\rho^{*} e^{*} = \sum_{\alpha} g_{\alpha}^{eq} (\mathbf{r} - \mathbf{e}_{\alpha} \delta_{t}, t - \delta_{t}), \tag{29}$$

where the superscript * indicates the intermediate quantities. To mathematically prove their consistency to the original macroscopic equations, we first apply the Taylor series expansion analysis to the equilibrium distribution functions with third-order truncated errors in space and time,

$$\begin{aligned} f_{\alpha}^{eq} (\mathbf{r} - \mathbf{e}_{\alpha} \delta_{t}, t - \delta_{t}) &= f_{\alpha}^{eq} (\mathbf{r}, t) - \delta_{t} \mathrm{D} f_{\alpha}^{eq} (\mathbf{r}, t) \\ &\quad + \frac{\delta_{t}^{2}}{2} \mathrm{D}^{2} f_{\alpha}^{eq} (\mathbf{r}, t) + O\left(\delta_{t}^{3} \right) \\ &= f_{\alpha}^{eq} (\mathbf{r}, t) - \delta_{t} \mathrm{D} f_{\alpha}^{eq} (\mathbf{r}, t) \\ &\quad - \frac{\delta_{t}}{2\tau_{\upsilon}} \mathrm{D} f_{\alpha}^{neq} (\mathbf{r}, t) + O\left(\delta_{t}^{3} \right), \end{aligned} \tag{30}$$

$$\begin{aligned} g_{\alpha}^{eq} (\mathbf{r} - \mathbf{e}_{\alpha} \delta_{t}, t - \delta_{t}) &= g_{\alpha}^{eq} (\mathbf{r}, t) - \delta_{t} \mathrm{D} g_{\alpha}^{eq} (\mathbf{r}, t) \\ &\quad + \frac{\delta_{t}^{2}}{2} \mathrm{D}^{2} g_{\alpha}^{eq} (\mathbf{r}, t) + O\left(\delta_{t}^{3} \right) \\ &= g_{\alpha}^{eq} (\mathbf{r}, t) - \delta_{t} \mathrm{D} g_{\alpha}^{eq} (\mathbf{r}, t) \\ &\quad - \frac{\delta_{t}}{2\tau_{c}} \mathrm{D} g_{\alpha}^{neq} (\mathbf{r}, t) + O\left(\delta_{t}^{3} \right). \end{aligned} \tag{31}$$

It should be noted that relationships given by the C-E analysis (Eqs. (17) and (18)) are adopted to derive the above formulations. By substituting Eqs. (30) and (31) into Eqs. (27)–(29) and performing certain mathematical manipulations, it can be shown that Eqs. (27)–(29) recover macroscopic equations (21)–(23) in the predictor step to second-order accuracy in space and time. For illustration purposes, substituting Eq. (30) into Eq. (27) yields

$$\begin{aligned} \rho^{*} = \sum_{\alpha} f_{\alpha}^{eq} (\mathbf{r}, t) - \delta_{t} \left[\frac{\partial}{\partial t} \sum_{\alpha} f_{\alpha}^{eq} (\mathbf{r}, t) + \nabla \cdot \sum_{\alpha} \mathbf{e}_{\alpha} f_{\alpha}^{eq} (\mathbf{r}, t) \right] \\ - \frac{\delta_{t}}{2\tau_{\upsilon}} \left[\frac{\partial}{\partial t} \sum_{\alpha} f_{\alpha}^{neq} (\mathbf{r}, t) + \nabla \cdot \sum_{\alpha} \mathbf{e}_{\alpha} f_{\alpha}^{neq} (\mathbf{r}, t) \right] + O(\delta_{t}^{3}). \end{aligned} \tag{32}$$

According to conservation laws, the first term on the Right-Hand-Side (RHS) is canceled by the term on Left-Hand-Side (LHS), and the third term on RHS vanishes. Therefore, the above equation can be simplified to

$$\frac{\partial}{\partial t} \rho (\mathbf{r}, t) + \nabla \cdot \sum_{\alpha} \mathbf{e}_{\alpha} f_{\alpha}^{eq} (\mathbf{r}, t) + O(\delta_{t}^{2}) = 0. \tag{33}$$

Eq. (27) is equivalent to Eq. (21) with second-order accuracy in both time and space, and the reconstruction of the solution to the intermediate density in the predictor step is thus accomplished. By performing similar manipulations on Eqs. (28) and (29), we can show that they recover solutions to Eqs. (22) and (23), respectively, with the second-order of accuracy in space and time.

In the corrector step, the formulations of USLBM are given as

$$\rho (\mathbf{r}, t) = \rho^{*}, \tag{34}$$

$$\begin{aligned} &\rho (\mathbf{r}, t) \, \mathbf{u} (\mathbf{r}, t) \\ &= \rho^{*} \mathbf{u}^{*} - \left(1 - \frac{1}{\tau_{\upsilon}} \right) \sum_{\alpha} \mathbf{e}_{\alpha} \left[f_{\alpha}^{neq} \left(\mathbf{r} + \frac{1}{2} \mathbf{e}_{\alpha} \delta_{t}, t - \frac{1}{2} \delta_{t} \right) \right. \\ &\quad \left. - f_{\alpha}^{neq} \left(\mathbf{r} - \frac{1}{2} \mathbf{e}_{\alpha} \delta_{t}, t - \frac{1}{2} \delta_{t} \right) \right] + \mathbf{F}_{E} \delta_{t}, \end{aligned} \tag{35}$$

$$\rho(\mathbf{r},t)\,e(\mathbf{r},t) = \rho^* e^* - \left(1-\frac{1}{\tau_c}\right)\sum_\alpha \left[g_\alpha^{neq}\left(\mathbf{r}+\frac{1}{2}\mathbf{e}_\alpha\delta_t, t-\frac{1}{2}\delta_t\right) - g_\alpha^{neq}\left(\mathbf{r}-\frac{1}{2}\mathbf{e}_\alpha\delta_t, t-\frac{1}{2}\delta_t\right)\right], \quad (36)$$

where the non-equilibrium distribution functions are evaluated by

$$f_\alpha^{neq}\left(\mathbf{r}-\frac{1}{2}\mathbf{e}_\alpha\delta_t, t-\frac{1}{2}\delta_t\right) = -\tau_\upsilon\left[f_\alpha^{eq}(\boldsymbol{r},t) - f_\alpha^{eq}(\mathbf{r}-\mathbf{e}_\alpha\delta_t, t-\delta_t)\right], \quad (37)$$

$$g_\alpha^{neq}\left(\mathbf{r}-\frac{1}{2}\mathbf{e}_\alpha\delta_t, t-\frac{1}{2}\delta_t\right) = -\tau_c\left[g_\alpha^{eq}(\mathbf{r},t) - g_\alpha^{eq}(\mathbf{r}-\mathbf{e}_\alpha\delta_t, t-\delta_t)\right]. \quad (38)$$

As per the suggestion of Shu *et al.*,[51] the first terms on the RHS of Eqs. (37) and (38) are evaluated by intermediate (denoted by asterisk) macroscopic quantities obtained in the predictor step, i.e., $f_\alpha^{eq}(\boldsymbol{r},t) = f_\alpha^{eq*}(\boldsymbol{r},t)$ and $g_\alpha^{eq}(\boldsymbol{r},t) = g_\alpha^{eq*}(\boldsymbol{r},t)$.

Clearly, Eq. (34) recovers Eq. (24) exactly. For a third-order truncated error, performing the Taylor series expansion analysis to Eqs. (37) and (38) yields

$$\begin{aligned}\tau_\upsilon\left[f_\alpha^{eq}(\boldsymbol{r},t) - f_\alpha^{eq}(\mathbf{r}-\mathbf{e}_\alpha\delta_t, t-\delta_t)\right] &= -\tau_\upsilon\delta_t \mathrm{D} f_\alpha^{eq}\left(\mathbf{r}-\frac{1}{2}\mathbf{e}_\alpha\delta_t, t-\frac{1}{2}\delta_t\right) + O\left(\delta_t^3\right)\\ &= f_\alpha^{neq}\left(\mathbf{r}-\frac{1}{2}\mathbf{e}_\alpha\delta_t, t-\frac{1}{2}\delta_t\right) + O\left(\delta_t^3\right), \end{aligned} \quad (39)$$

$$\begin{aligned}\tau_c\left[g_\alpha^{eq}(\boldsymbol{r},t) - g_\alpha^{eq}(\mathbf{r}-\mathbf{e}_\alpha\delta_t, t-\delta_t)\right] &= -\tau_c\delta_t \mathrm{D} g_\alpha^{eq}\left(\mathbf{r}-\frac{1}{2}\mathbf{e}_\alpha\delta_t, t-\frac{1}{2}\delta_t\right) + O\left(\delta_t^3\right)\\ &= g_\alpha^{neq}\left(\mathbf{r}-\frac{1}{2}\mathbf{e}_\alpha\delta_t, t-\frac{1}{2}\delta_t\right) + O\left(\delta_t^3\right). \end{aligned} \quad (40)$$

The above formulations can be further expanded with respect to the time/space level of $(\mathbf{r},\ t-0.5\delta_t)$, which gives

$$\begin{aligned} f_\alpha^{neq}&\left(\mathbf{r}\pm\frac{1}{2}\mathbf{e}_\alpha\delta_t, t-\frac{1}{2}\delta_t\right)\\ &= f_\alpha^{neq}\left(\mathbf{r}, t-\frac{1}{2}\delta_t\right) \pm \left(\frac{1}{2}\delta_t\mathbf{e}_\alpha\cdot\nabla\right) f_\alpha^{neq}\left(\mathbf{r}, t-\frac{1}{2}\delta_t\right)\\ &\quad + \frac{1}{2}\left(\frac{1}{2}\delta_t\mathbf{e}_\alpha\cdot\nabla\right)^2 f_\alpha^{neq}\left(\mathbf{r}, t-\frac{1}{2}\delta_t\right) + O\left(\delta_t^3\right), \end{aligned} \quad (41)$$

$$\begin{aligned} g_\alpha^{neq}&\left(\mathbf{r}\pm\frac{1}{2}\mathbf{e}_\alpha\delta_t, t-\frac{1}{2}\delta_t\right)\\ &= g_\alpha^{neq}\left(\mathbf{r}, t-\frac{1}{2}\delta_t\right) \pm \left(\frac{1}{2}\delta_t\mathbf{e}_\alpha\cdot\nabla\right) g_\alpha^{neq}\left(\mathbf{r}, t-\frac{1}{2}\delta_t\right)\\ &\quad + \frac{1}{2}\left(\frac{1}{2}\delta_t\mathbf{e}_\alpha\cdot\nabla\right)^2 g_\alpha^{neq}\left(\mathbf{r}, t-\frac{1}{2}\delta_t\right) + O\left(\delta_t^3\right). \end{aligned} \quad (42)$$

Substituting the above expansion formulations into the second terms on RHS in Eqs. (35) and (36) leads to

$$\begin{aligned} -\left(1-\frac{1}{\tau_\upsilon}\right)\frac{1}{\delta_t}\sum_\alpha \mathbf{e}_\alpha &\left[f_\alpha^{neq}\left(\mathbf{r}+\frac{1}{2}\mathbf{e}_\alpha\delta_t, t-\frac{1}{2}\delta_t\right) - f_\alpha^{neq}\left(\mathbf{r}-\frac{1}{2}\mathbf{e}_\alpha\delta_t, t-\frac{1}{2}\delta_t\right)\right]\\ &= -\nabla\cdot\left[\left(1-\frac{1}{\tau_\upsilon}\right)\sum_\alpha (\mathbf{e}_\alpha)_\beta(\mathbf{e}_\alpha)_\gamma f_\alpha^{neq}\left(\mathbf{r}, t-\frac{1}{2}\delta_t\right)\right] + O\left(\delta_t^2\right), \end{aligned} \quad (43)$$

$$\begin{aligned} -\left(1-\frac{1}{\tau_c}\right)\frac{1}{\delta_t}\sum_\alpha &\left[g_\alpha^{neq}\left(\mathbf{r}+\frac{1}{2}\mathbf{e}_\alpha\delta_t, t-\frac{1}{2}\delta_t\right) - g_\alpha^{neq}\left(\mathbf{r}-\frac{1}{2}\mathbf{e}_\alpha\delta_t, t-\frac{1}{2}\delta_t\right)\right]\\ &= -\nabla\cdot\left[\left(1-\frac{1}{\tau_c}\right)\sum_\alpha \mathbf{e}_\alpha g_\alpha^{neq}\left(\mathbf{r}, t-\frac{1}{2}\delta_t\right)\right] + O\left(\delta_t^2\right). \end{aligned} \quad (44)$$

As can be seen, Eqs. (43) and (44) recover the spatial derivative terms in Eqs. (25) and (26), respectively, with second-order accuracy in space. It is also noted that the approximated quantities are at the time level of $(t-0.5\delta_t)$, which represents a typical central difference scheme that ensures the second-order of accuracy in time.

The remaining part in the corrector step is the external force term $\mathbf{F}_E$ in velocity correction. As shown in Eq. (4), no spatial derivative is involved in this term. Therefore, it is unnecessary to consider the spatial accuracy when performing numerical approximations. To maintain the second-order accuracy in time, the central difference scheme is adopted, with which the external force term is approximated as

$$F_{Ez} = -\frac{g\beta}{2}\left[\rho(\mathbf{r},t-\delta_t)\left(T(\mathbf{r},t-\delta_t)-T_m\right) + \rho(\mathbf{r},t)\left(T(\mathbf{r},t)-T_m\right)\right]. \quad (45)$$

Note that $T(\mathbf{r},\ t)$ can be evaluated from the internal energy obtained in the corrector step. The above approximation of the external force term ensures the second-order accuracy in time, and at the same time, maintains the explicitness of the present solver. Therefore, it is proven that Eqs. (35) and (36) recover the macroscopic equations (25) and (26) with the second-order of accuracy in space and time.

Further simplification can be made to Eqs. (35) and (36). By using formulations (28) and (29) in the predictor step, the following relationships for/between the calculated non-equilibrium distribution functions can be derived:

$$\begin{aligned}\sum_{\alpha}\mathbf{e}_{\alpha}f_{\alpha}^{neq}\left(\mathbf{r}-\frac{1}{2}\mathbf{e}_{\alpha}\delta_t,t-\frac{1}{2}\delta_t\right)&=-\tau_{\upsilon}\left[\sum_{\alpha}\mathbf{e}_{\alpha}f_{\alpha}^{eq*}(\mathbf{r},t)-\sum_{\alpha}\mathbf{e}_{\alpha}f_{\alpha}^{eq}(\mathbf{r}-\mathbf{e}_{\alpha}\delta_t,t-\delta_t)\right]\\&=-\tau_{\upsilon}\left[\rho^{*}\mathbf{u}^{*}-\rho^{*}\mathbf{u}^{*}\right]\\&=0,\end{aligned}\tag{46}$$

$$\begin{aligned}\sum_{\alpha}g_{\alpha}^{neq}\left(\mathbf{r}-\frac{1}{2}\mathbf{e}_{\alpha}\delta_t,t-\frac{1}{2}\delta_t\right)&=-\tau_{c}\left[\sum_{\alpha}g_{\alpha}^{eq*}(\mathbf{r},t)-\sum_{\alpha}g_{\alpha}^{eq}(\mathbf{r}-\mathbf{e}_{\alpha}\delta_t,t-\delta_t)\right]\\&=-\tau_{c}\left[\rho^{*}e^{*}-\rho^{*}e^{*}\right]\\&=0.\end{aligned}\tag{47}$$

With the above relationships, formulations (35) and (36) of the corrector step can be further reduced to

$$\rho(\mathbf{r},t)\,\mathbf{u}(\mathbf{r},t)=\rho^{*}\mathbf{u}^{*}-\left(1-\frac{1}{\tau_{\upsilon}}\right)\sum_{\alpha}\mathbf{e}_{\alpha}f_{\alpha}^{neq}\times\left(\mathbf{r}+\frac{1}{2}\mathbf{e}_{\alpha}\delta_t,t-\frac{1}{2}\delta_t\right)+\mathbf{F}_{E}\delta_t,\tag{48}$$

$$\rho(\mathbf{r},t)\,e(\mathbf{r},t)=\rho^{*}e^{*}-\left(1-\frac{1}{\tau_{c}}\right)\sum_{\alpha}g_{\alpha}^{neq}\times\left(\mathbf{r}+\frac{1}{2}\mathbf{e}_{\alpha}\delta_t,t-\frac{1}{2}\delta_t\right).\tag{49}$$

To sum up, the formulations of the present 3D-USLBM for incompressible isothermal/thermal flows can be expressed as follows.

Predictor step:

$$\rho^{*}=\sum_{\alpha}f_{\alpha}^{eq}(\mathbf{r}-\mathbf{e}_{\alpha}\delta_t,t-\delta_t),\tag{50}$$

$$\rho^{*}\mathbf{u}^{*}=\sum_{\alpha}\mathbf{e}_{\alpha}f_{\alpha}^{eq}(\mathbf{r}-\mathbf{e}_{\alpha}\delta_t,t-\delta_t),\tag{51}$$

$$\rho^{*}e^{*}=\sum_{\alpha}g_{\alpha}^{eq}(\mathbf{r}-\mathbf{e}_{\alpha}\delta_t,t-\delta_t).\tag{52}$$

Corrector step:

$$\rho(\mathbf{r},t)=\rho^{*},\tag{53}$$

$$\rho(\mathbf{r},t)\,\mathbf{u}(\mathbf{r},t)=\rho^{*}\mathbf{u}^{*}-\left(1-\frac{1}{\tau_{\upsilon}}\right)\sum_{\alpha}\mathbf{e}_{\alpha}f_{\alpha}^{neq}\times\left(\mathbf{r}+\frac{1}{2}\mathbf{e}_{\alpha}\delta_t,t-\frac{1}{2}\delta_t\right)+\mathbf{F}_{E}\delta_t,\tag{54}$$

$$\rho(\mathbf{r},t)\,e(\mathbf{r},t)=\rho^{*}e^{*}-\left(1-\frac{1}{\tau_{c}}\right)\sum_{\alpha}g_{\alpha}^{neq}\times\left(\mathbf{r}+\frac{1}{2}\mathbf{e}_{\alpha}\delta_t,t-\frac{1}{2}\delta_t\right),\tag{55}$$

where the non-equilibrium terms are evaluated by Eqs. (37) and (38), and the external force term is approximated by Eq. (45).

It is noteworthy that in 3D-USLBM, the final formulations only involve the equilibrium distribution functions and the non-equilibrium distribution functions. Specifically, the equilibrium distribution function is evaluated from the macroscopic variables, while the non-equilibrium distribution function is approximated by the difference between two equilibrium distribution functions at different locations and time levels. Therefore, compared to the standard three-dimensional lattice Boltzmann method (3D LBM), 3D-USLBM gives remarkable savings in virtual memory. Take the widely used D3Q15 model as an example. In the standard 3D LBM, on each grid point, it is necessary to store at least 15 distribution functions in isothermal simulations and at least 30 distribution functions in thermal problems. Comparatively, in the 3D-USLBM, even if we store the macroscopic variables at two time levels, on each grid point, only 8 variables in isothermal flows and 10 variables in thermal simulations are required. So theoretically, around 50% reduction in virtual memory can be expected in 3D-USLBM simulations.

Apart from the saving in virtual memory, the 3D-USLBM also facilitates the implementation of physical boundary conditions and shows better numerical stability as compared to the standard LBM. These merits will be discussed in detail in Sections III B and III C.

B. Boundary conditions

In the standard LBM, the physical boundary conditions related to the macroscopic variables need to be transformed into the boundary conditions of the distribution functions so as to be implemented on the boundary points. Such transformations are usually tedious and complicated, especially for 3D problems. In the 3D-USLBM, however, the macroscopic variables, rather than the distribution functions, are directly updated. As a result, the physical boundary conditions can be directly implemented in 3D-USLBM without any transformation.

Another issue of the boundary treatment in 3D-USLBM is the evaluation of the non-equilibrium distribution functions at the boundary points which are involved in the calculation of the corrector step. One popular boundary treatment of the non-equilibrium distribution function is the non-equilibrium extrapolation scheme[29] which ensures the second-order accuracy in space and can be flexibly incorporated into non-uniform meshes or problems with complex boundaries. In such treatments, the boundary values of non-equilibrium distribution functions are set equal to those on the inner layer of mesh points.

C. Stability analysis

Numerical stability is one of the main aspects in assessing a new method. In this section, the von Neumann stability analysis is conducted on the present 3D-USLBM to evaluate its stability. For the ease of analysis, the following variables are defined:

$$y_1 = \rho, \quad y_2 = \rho u, \quad y_3 = \rho v, \quad y_4 = \rho w, \quad y_5 = \rho e, \tag{56}$$

and the variable vector is

$$\mathbf{Y} = (y_1 \;\; y_2 \;\; y_3 \;\; y_4 \;\; y_5)^T. \tag{57}$$

Substituting the notation into the equilibrium distribution functions (9) and (10) gives

$$\begin{aligned} f_\alpha^{eq} = {} & w_\alpha y_{1\alpha} + \frac{w_\alpha}{c_s^2}\left(e_{\alpha x} y_{2\alpha} + e_{\alpha y} y_{3\alpha} + e_{\alpha z} y_{4\alpha}\right) \\ & + \frac{w_\alpha}{2c_s^4}\frac{\left(e_{\alpha x} y_{2\alpha} + e_{\alpha y} y_{3\alpha} + e_{\alpha z} y_{4\alpha}\right)^2}{y_{1\alpha}} \\ & - \frac{w_\alpha}{2c_s^2}\frac{y_{2\alpha}^2 + y_{3\alpha}^2 + y_{4\alpha}^2}{y_{1\alpha}}, \end{aligned} \tag{58}$$

$$g_\alpha^{eq} = \begin{cases} -\dfrac{1}{3} y_{5\alpha}\dfrac{y_{2\alpha}^2 + y_{3\alpha}^2 + y_{4\alpha}^2}{y_{1\alpha}^2}, & \alpha = 0 \\ \dfrac{1}{9} y_{5\alpha} + \dfrac{1}{9} y_{5\alpha}\dfrac{e_{\alpha x} y_{2\alpha} + e_{\alpha y} y_{3\alpha} + e_{\alpha z} y_{4\alpha}}{y_{1\alpha}} + \dfrac{1}{2} y_{5\alpha}\left(\dfrac{e_{\alpha x} y_{2\alpha} + e_{\alpha y} y_{3\alpha} + e_{\alpha z} y_{4\alpha}}{y_{1\alpha}}\right)^2 & \\ -\dfrac{1}{6} y_{5\alpha}\dfrac{y_{2\alpha}^2 + y_{3\alpha}^2 + y_{4\alpha}^2}{y_{1\alpha}^2}, & \alpha = 1 \sim 6 \\ \dfrac{1}{24} y_{5\alpha} + \dfrac{7}{72} y_{5\alpha}\dfrac{e_{\alpha x} y_{2\alpha} + e_{\alpha y} y_{3\alpha} + e_{\alpha z} y_{4\alpha}}{y_{1\alpha}} + \dfrac{1}{16} y_{5\alpha}\left(\dfrac{e_{\alpha x} y_{2\alpha} + e_{\alpha y} y_{3\alpha} + e_{\alpha z} y_{4\alpha}}{y_{1\alpha}}\right)^2 & \\ -\dfrac{1}{48} y_{5\alpha}\dfrac{y_{2\alpha}^2 + y_{3\alpha}^2 + y_{4\alpha}^2}{y_{1\alpha}^2}, & \alpha = 7 \sim 14 \end{cases}, \tag{59}$$

where the subscript α denotes the quantities at a space level of $(\mathbf{r} - \mathbf{e}_\alpha \delta_t)$. The above formulations can be substituted into Eqs. (50)–(52) in the predictor step, which results in a system of equations denoted as

$$\mathbf{Y}^* = \mathbf{G}\left(\mathbf{Y}^n\right), \tag{60}$$

where the superscript n represents the time level. Note that Eqs. (58) and (59) involve nonlinear terms, therefore the resultant system of equations (60) is also nonlinear. To perform the von Neumann stability analysis which is only applicable to linear systems, a linearization process is introduced to Eq. (60), through which the system of equations (60) is differentiated,

$$\begin{aligned} \mathrm{d}\mathbf{Y}^* = {} & \left[\frac{\partial \mathbf{Y}^*}{\partial y_1}\right]^n \mathrm{d}y_1^n + \left[\frac{\partial \mathbf{Y}^*}{\partial y_2}\right]^n \mathrm{d}y_2^n + \left[\frac{\partial \mathbf{Y}^*}{\partial y_3}\right]^n \mathrm{d}y_3^n \\ & + \left[\frac{\partial \mathbf{Y}^*}{\partial y_4}\right]^n \mathrm{d}y_4^n + \left[\frac{\partial \mathbf{Y}^*}{\partial y_5}\right]^n \mathrm{d}y_5^n. \end{aligned} \tag{61}$$

In Eq. (61), all terms inside the brackets are based on the known values at the time level n. Therefore, it expresses a linear relationship between $\mathrm{d}\mathbf{Y}^*$ and $\mathrm{d}y_1^n$, $\mathrm{d}y_2^n$, $\mathrm{d}y_3^n$, $\mathrm{d}y_4^n$, $\mathrm{d}y_5^n$. For instance, the differentiation of y_1^* can be written as

$$\mathrm{d}y_1^* = \left(\frac{\partial y_1^*}{\partial y_1}\right)^n \mathrm{d}y_1^n + \left(\frac{\partial y_1^*}{\partial y_2}\right)^n \mathrm{d}y_2^n + \left(\frac{\partial y_1^*}{\partial y_3}\right)^n \mathrm{d}y_3^n + \left(\frac{\partial y_1^*}{\partial y_4}\right)^n \mathrm{d}y_4^n. \tag{62}$$

The parameter of the first term on the right-hand-side is chosen to illustrate our analysis, which has the following form:

$$\begin{aligned} \left(\frac{\partial y_1^*}{\partial y_1}\right)^n = {} & \sum_\alpha w_\alpha - \sum_\alpha \frac{w_\alpha}{2c_s^4\left(y_{1\alpha}^n\right)^2}\left(e_{\alpha x} y_{2\alpha}^n + e_{\alpha y} y_{3\alpha}^n + e_{\alpha z} y_{4\alpha}^n\right)^2 \\ & + \sum_\alpha \frac{w_\alpha}{2c_s^2\left(y_{1\alpha}^n\right)^2}\left(\left(y_{2\alpha}^n\right)^2 + \left(y_{3\alpha}^n\right)^2 + \left(y_{4\alpha}^n\right)^2\right). \end{aligned} \tag{63}$$

To apply the von-Neumann stability analysis,[4] it is assumed that the variables $\mathrm{d}y_i^*, \mathrm{d}y_i^n, y_{i\alpha}^n$ can be approximated by the Fourier series expansion. Consider a random component of these variables, which can be written as

$$dy_i^* = \overline{dy}_i^* e^{j\mathbf{k}\cdot\mathbf{r}}, \quad dy_i^n = \overline{dy}_i^n e^{j\mathbf{k}\cdot\mathbf{r}}, \quad y_{i\alpha}^n = y_i^n e^{j\mathbf{k}\cdot(\mathbf{r}-\mathbf{e}_\alpha\delta_t)}, \quad i = 1, 2, 3. \tag{64}$$

Substituting the above notation into Eq. (63) gives

$$\begin{aligned} \left(\frac{\partial y_1^*}{\partial y_1}\right)^n = {} & \sum_\alpha w_\alpha - \sum_\alpha \frac{w_\alpha}{2c_s^4}\left(\frac{e_{\alpha x} y_2^n + e_{\alpha y} y_3^n + e_{\alpha z} y_4^n}{y_1^n}\right)^2 \\ & + \sum_\alpha \frac{w_\alpha}{2c_s^2}\frac{\left(y_2^n\right)^2 + \left(y_3^n\right)^2 + \left(y_4^n\right)^2}{\left(y_1^n\right)^2}. \end{aligned} \tag{65}$$

Note that for the D3Q15 model, the following isentropic properties of lattice tensors are available:

$$\begin{aligned} & \sum_\alpha w_\alpha e_{\alpha x}^2 = \sum_\alpha w_\alpha e_{\alpha y}^2 = \sum_\alpha w_\alpha e_{\alpha z}^2 = c_s^2, \\ & \sum_\alpha w_\alpha e_{\alpha x} e_{\alpha y} = \sum_\alpha w_\alpha e_{\alpha y} e_{\alpha z} = \sum_\alpha w_\alpha e_{\alpha z} e_{\alpha x} = 0, \\ & \sum_\alpha w_\alpha = 1, \\ & \sum_\alpha w_\alpha e_{\alpha x} = \sum_\alpha w_\alpha e_{\alpha y} = \sum_\alpha w_\alpha e_{\alpha z} = 0. \end{aligned} \tag{66}$$

With these relationships, Eq. (65) can be reduced further to

$$\left(\frac{\partial y_1^*}{\partial y_1}\right)^n = 1. \tag{67}$$

Through a similar process, it can be proven that other parameters obey the following relationship:

$$\left(\frac{\partial y_i^*}{\partial y_j}\right)^n = \delta_{ij}. \tag{68}$$

Subsequently, Eq. (61) is simplified to

$$\overline{\mathrm{d}\mathbf{Y}}^{*} = \overline{\mathrm{d}\mathbf{Y}}^{n}. \tag{69}$$

In the corrector step, no correction on the density is conducted. Therefore by substituting $\mathrm{d}y_i^{n+1} = \overline{\mathrm{d}y_i}^{n+1} e^{j\mathbf{k}\cdot\mathbf{r}}$ and adopting the relationship given in Eq. (69), we have

$$\overline{\mathrm{d}y_1}^{n+1} = \overline{\mathrm{d}y_1}^{n}. \tag{70}$$

According to Eqs. (37), (38), (54), and (55) and by adopting the formulations (50)–(52) in the predictor step, the corrections for the density and inner energy are

$$y_2^{n+1} = y_2^{*} + (\tau_\upsilon - 1)\sum_\alpha e_{\alpha x} f_\alpha^{eq}(\mathbf{r} + \mathbf{e}_\alpha\delta_t, t) - (\tau_\upsilon - 1)\, y_2^{n}, \tag{71}$$

$$y_3^{n+1} = y_3^{*} + (\tau_\upsilon - 1)\sum_\alpha e_{\alpha y} f_\alpha^{eq}(\mathbf{r} + \mathbf{e}_\alpha\delta_t, t) - (\tau_\upsilon - 1)\, y_3^{n}, \tag{72}$$

$$\begin{aligned} y_4^{n+1} &= y_4^{*} + (\tau_\upsilon - 1)\sum_\alpha e_{\alpha z} f_\alpha^{eq}(\mathbf{r} + \mathbf{e}_\alpha\delta_t, t) - (\tau_\upsilon - 1)\, y_4^{n} \\ &\quad - \frac{g\beta}{2}\left[\frac{2}{3}y_5^{n} + \frac{2}{3}y_5^{n+1} - \left(y_1^{n} + y_1^{n+1}\right) T_m\right]\delta_t, \end{aligned} \tag{73}$$

$$y_5^{n+1} = y_5^{*} + (\tau_c - 1)\sum_\alpha g_\alpha^{eq}(\mathbf{r} + \mathbf{e}_\alpha\delta_t, t) - (\tau_c - 1)\, y_5^{n}. \tag{74}$$

For illustration purposes, we perform the stability analysis on Eq. (71). The second term on its RHS can be written as

$$\begin{aligned} S &= \sum_\alpha e_{\alpha x} f_\alpha^{eq}(\mathbf{r} + \mathbf{e}_\alpha\delta_t, t) \\ &= \sum_\alpha e_{\alpha x} w_\alpha y_{1-\alpha}^{*} + \sum_\alpha e_{\alpha x}\frac{w_\alpha}{c_s^2}\left(e_{\alpha x}y_{2-\alpha}^{*} + e_{\alpha y}y_{3-\alpha}^{*} + e_{\alpha z}y_{4-\alpha}^{*}\right) \\ &\quad + \sum_\alpha e_{\alpha x}\frac{w_\alpha}{2c_s^4}\frac{\left(e_{\alpha x}y_{2-\alpha}^{*} + e_{\alpha y}y_{3-\alpha}^{*} + e_{\alpha z}y_{4-\alpha}^{*}\right)^2}{y_{1-\alpha}^{*}} \\ &\quad - \sum_\alpha e_{\alpha x}\frac{w_\alpha}{2c_s^2}\frac{\left(y_{2-\alpha}^{*}\right)^2 + \left(y_{3-\alpha}^{*}\right)^2 + \left(y_{4-\alpha}^{*}\right)^2}{y_{1-\alpha}^{*}}, \end{aligned} \tag{75}$$

where the subscript "$-\alpha$" means the quantities at a space level of $(\mathbf{r} + \mathbf{e}_\alpha\delta_t)$. The term S is linearized by differentiation,

$$\mathrm{d}S = \left(\frac{\partial S}{\partial y_1}\right)^{*}\mathrm{d}y_1^{*} + \left(\frac{\partial S}{\partial y_2}\right)^{*}\mathrm{d}y_2^{*} + \left(\frac{\partial S}{\partial y_3}\right)^{*}\mathrm{d}y_3^{*} + \left(\frac{\partial S}{\partial y_4}\right)^{*}\mathrm{d}y_4^{*}. \tag{76}$$

Specifically, the parameter of the first term on RHS is

$$\begin{aligned} \left(\frac{\partial S}{\partial y_1}\right)^{*} &= \sum_\alpha e_{\alpha x} w_\alpha \\ &\quad - \sum_\alpha e_{\alpha x}\frac{w_\alpha}{2c_s^4}\frac{\left(e_{\alpha x}y_{2-\alpha}^{*} + e_{\alpha y}y_{3-\alpha}^{*} + e_{\alpha z}y_{4-\alpha}^{*}\right)^2}{\left(y_{1-\alpha}^{*}\right)^2} \\ &\quad + \sum_\alpha e_{\alpha x}\frac{w_\alpha}{2c_s^2}\frac{\left(y_{2-\alpha}^{*}\right)^2 + \left(y_{3-\alpha}^{*}\right)^2 + \left(y_{4-\alpha}^{*}\right)^2}{\left(y_{1-\alpha}^{*}\right)^2}. \end{aligned} \tag{77}$$

By substituting in Eq. (64) and the isentropic properties (66), the above equation can be simplified to

$$\begin{aligned} \left(\frac{\partial S}{\partial y_1}\right)^{*} &= \sum_\alpha e_{\alpha x} w_\alpha - \sum_\alpha e_{\alpha x}\frac{w_\alpha}{2c_s^4}\left(\frac{e_{\alpha x}y_2^{*} + e_{\alpha y}y_3^{*} + e_{\alpha z}y_4^{*}}{y_1^{*}}\right)^2 \\ &\quad + \sum_\alpha e_{\alpha x}\frac{w_\alpha}{2c_s^2}\frac{\left(y_2^{*}\right)^2 + \left(y_3^{*}\right)^2 + \left(y_4^{*}\right)^2}{\left(y_1^{*}\right)^2} \\ &= H_1 - H_2 + H_3, \end{aligned} \tag{78}$$

where

$$H_1 = \sum_\alpha w_\alpha e_{\alpha x} = 0, \tag{79}$$

$$\begin{aligned} H_2 &= \frac{\left(y_2^{*}\right)^2}{2c_s^4\left(y_1^{*}\right)^2}\sum_\alpha w_\alpha e_{\alpha x}^3 + \frac{\left(y_3^{*}\right)^2}{2c_s^4\left(y_1^{*}\right)^2}\sum_\alpha w_\alpha e_{\alpha x}e_{\alpha y}^2 \\ &\quad + \frac{\left(y_4^{*}\right)^2}{2c_s^4\left(y_1^{*}\right)^2}\sum_\alpha w_\alpha e_{\alpha x}e_{\alpha z}^2 + \frac{y_2^{*}y_3^{*}}{c_s^4\left(y_1^{*}\right)^2}\sum_\alpha w_\alpha e_{\alpha x}^2 e_{\alpha y} \\ &\quad + \frac{y_2^{*}y_4^{*}}{c_s^4\left(y_1^{*}\right)^2}\sum_\alpha w_\alpha e_{\alpha x}^2 e_{\alpha z} + \frac{y_3^{*}y_4^{*}}{c_s^4\left(y_1^{*}\right)^2}\sum_\alpha w_\alpha e_{\alpha x}e_{\alpha y}e_{\alpha z} \\ &= 0, \end{aligned} \tag{80}$$

$$H_3 = \frac{\left(y_2^{*}\right)^2 + \left(y_3^{*}\right)^2 + \left(y_4^{*}\right)^2}{2c_s^2\left(y_1^{*}\right)^2}\sum_\alpha w_\alpha e_{\alpha x} = 0. \tag{81}$$

With the above formulations, we can derive

$$\left(\frac{\partial S}{\partial y_1}\right)^{*} = 0. \tag{82}$$

Following a similar process for the other RHS terms in Eq. (76), the following relationships can be obtained:

$$\left(\frac{\partial S}{\partial y_2}\right)^{*} = 1, \quad \left(\frac{\partial S}{\partial y_3}\right)^{*} = 0, \quad \left(\frac{\partial S}{\partial y_4}\right)^{*} = 0. \tag{83}$$

Together with Eqs. (64), (82), and (83), Eq. (76) can thus be simplified to

$$\overline{\mathrm{d}S} = \overline{\mathrm{d}y_2^{*}}. \tag{84}$$

Differentiating Eq. (71) and using Eqs. (69) and (84), we obtain

$$\overline{\mathrm{d}y_2}^{n+1} = \overline{\mathrm{d}y_2}^{n}. \tag{85}$$

Following a similar process as shown in Eqs. (75)–(85), the stability analysis on Eqs. (72)–(74) can be performed to derive the following relationships:

$$\begin{cases} \overline{\mathrm{d}y_1}^{n+1} = \overline{\mathrm{d}y_1}^{n} \\ \overline{\mathrm{d}y_2}^{n+1} = \overline{\mathrm{d}y_2}^{n} \\ \overline{\mathrm{d}y_3}^{n+1} = \overline{\mathrm{d}y_3}^{n} \\ \overline{\mathrm{d}y_4}^{n+1} = g\beta T_m\delta_t\overline{\mathrm{d}y_1}^{n} + \overline{\mathrm{d}y_4}^{n} - 2g\beta\delta_t\overline{\mathrm{d}y_5}^{n}/3 \\ \overline{\mathrm{d}y_5}^{n+1} = \overline{\mathrm{d}y_5}^{n} \end{cases}. \tag{86}$$

Finally, the stability analysis of 3D-USLBM can be written in the following matrix form

$$\overline{\mathrm{d}\mathbf{Y}}^{n+1} = \begin{bmatrix} 1 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 \\ g\beta T_m \delta_t & 0 & 0 & 1 & -2g\beta\delta_t/3 \\ 0 & 0 & 0 & 0 & 1 \end{bmatrix} \cdot \overline{\mathrm{d}\mathbf{Y}}^{n}. \quad (87)$$

Obviously, all eigenvalues of the characteristic matrix are 1. Therefore, the present 3D-USLBM is unconditionally stable.

D. Computational sequence

The computational sequence of the three-dimensional simplified and unconditionally stable thermal lattice Boltzmann method (3D-USLBM) can be summarized as follows:

i. Specify the streaming distance δ_x ($\delta_x = \delta_t$). Determine the single relaxation parameters τ_v and τ_c.
ii. Predictor step: Use Eqs. (50)–(52) to calculate the intermediate flow variables ρ^*, $\mathbf{u}^*$, and e^*.
iii. Compute the non-equilibrium distribution functions (f_α^{neq} and $g_\alpha{}^{neq}$) by applying Eqs. (37) and (38).
iv. Implement appropriate boundary conditions for the non-equilibrium distribution functions.
v. Corrector step: Obtain density, velocity, and internal energy at the new time step from Eqs. (53)–(55). The external force term $\mathbf{F}_E$ involved in this step can be evaluated from Eq. (45).
vi. Implement appropriate boundary conditions for the macroscopic flow variables.
vii. Repeat steps (ii)–(vi) until the computation converges or the prescribed maximum time step is reached.

IV. NUMERICAL RESULTS AND DISCUSSIONS

To comprehensively evaluate the performance of the present 3D-USLBM, four representative numerical examples, including plane Poiseuille flow, lid-driven cavity flow, natural convection in a cubic cavity, and natural convection in a concentric annulus, are presented in this section. Constrained by the incompressible limit, the Mach number (Ma) in all simulations is maintained at lower than 0.3. The D3Q15 lattice velocity model adopted in this paper gives the sound speed C_s as $1/\sqrt{3}$. Thus, Ma = $U/C_s < 0.3$ leads to $U < 0.3C_s = 0.173$ (U is the characteristic velocity). To fulfill that requirement, all characteristic velocities in our simulations are chosen as $U = 0.1$. The accuracy, solution convergence, and stability of 3D-USLBM as well as its flexibility in a non-uniform mesh and problems with curved boundaries are validated in these examples. In all tests, good agreements with the analytical results or numerical solutions are observed.

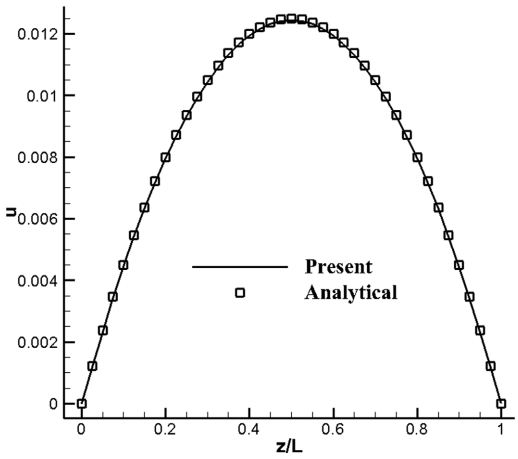


FIG. 2. Comparison of u-velocity distribution along the vertical centerline for plane Poiseuille flow.

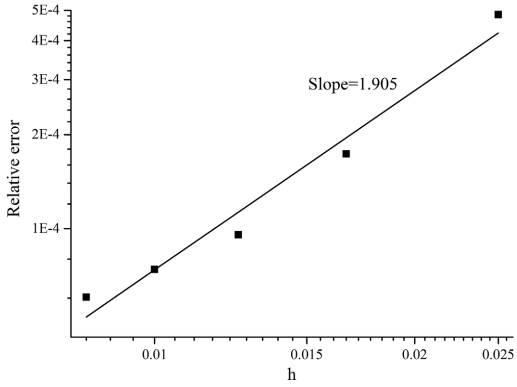


FIG. 3. L_2 norm of relative error of u versus mesh spacing h for plane Poiseuille flow.

A. Plane Poiseuille flow between two parallel plates

Plane Poiseuille flow is a classic benchmark test with analytical solutions available for reference. We choose this case to examine the numerical accuracy of 3D-USLBM. The physical configuration is the internal flow between two parallel plates, driven by a constant pressure gradient in the flow direction. The simulation domain adopted in our test is cuboid with dimensions $x \times y \times z = 0.5L \times 0.5L \times L$. A no-slip boundary condition is implemented on the upper and lower boundaries at $z = 0$ and $z = L$, respectively, while the periodic boundary condition is applied on boundaries in the y-direction, and the Dirichlet condition of pressure is adopted on boundaries in the x-direction to impose a constant pressure gradient. The analytical solution to the problem is

$$u(x,y,z) = \frac{1}{2\mu}\left(\frac{\partial p}{\partial x}\right)\left(z^2 - Lz\right), \quad v(x,y,z) = 0, \quad w(x,y,z) = 0. \quad (88)$$

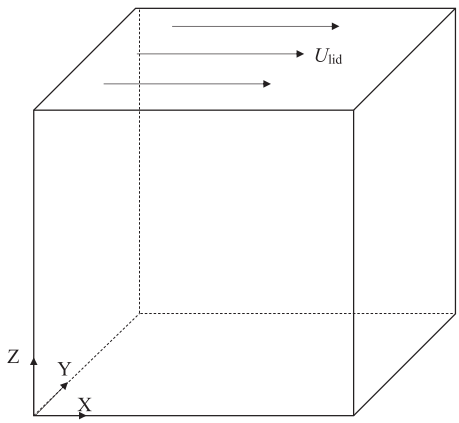


FIG. 4. Configuration of the 3D lid-driven cavity flow.

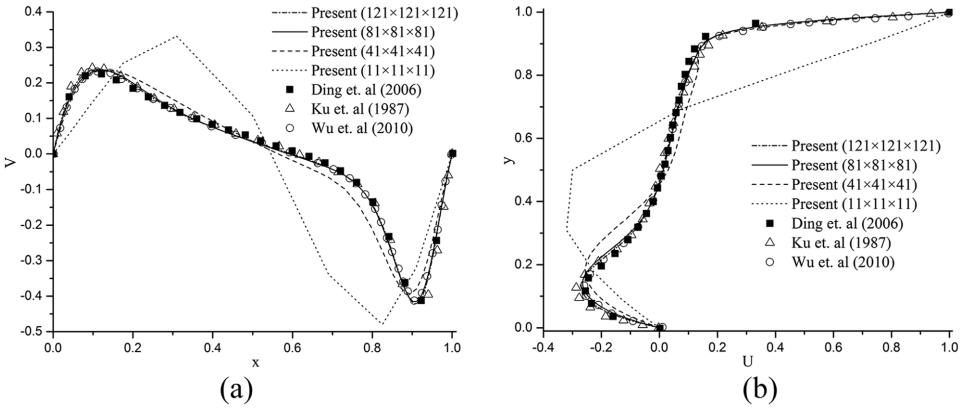


FIG. 5. Velocity distribution of the 3D lid-driven cavity flow with Re = 1000 at different mesh resolutions. Left: x-V curve; right: U-y curve.

In our simulations, the distance between two parallel plates is $L = 1$, and the pressure gradient $\partial p/\partial x$ and the dynamic viscosity μ are set to be −0.0001 and 0.001, respectively. For a given mesh spacing and streaming distance, the single relaxation parameter τ_υ can be obtained from Eq. (19).

Figure 2 presents the distribution of u-velocity along the vertical centerline of the computational domain, with a mesh size of $21 \times 21 \times 41$. It can be seen that the numerical results are in good agreement with the analytical solution. To further validate the spatial accuracy of the present method, additional cases with mesh sizes of $31 \times 31 \times 61$, $41 \times 41 \times 81$, $51 \times 51 \times 101$, and $61 \times 61 \times 121$ are simulated, and the L_2 norms of relative error of horizontal velocity u along the vertical centerline are calculated according to the following formulation:

$$L_2 = \sqrt{\frac{1}{N_{\text{total}}} \sum_{k=1}^{N_{\text{total}}} \left(\frac{u_{\text{numerical}} - u_{\text{exact}}}{U_0} \right)^2}, \tag{89}$$

where U_0 denotes the maximum velocity in the analytical solution. As can be seen in Figure 3, the slope of the linearly fitted curve of L_2 norm versus mesh spacing is 1.905. In numerical analysis, the order of accuracy quantifies the rate of convergence. Therefore, the convergence of 3D-USLBM is dominantly second order in space, although there are some effects arising from varying relaxation parameters (reflected as a minor nonlinear behavior in Figure 3).

B. Lid-driven cavity flow

Here, 3D lid-driven cavity flow (Figure 4) is simulated to examine the ability of the present method in modeling 3D incompressible viscous flows. The flow shapes of the present problem are affected by the Reynolds number (defined as $\text{Re} = U_{\text{lid}} L/\upsilon$, where U_{lid} is the moving velocity of the lid and L is the length of the cubic cavity). The flow parameters applied in the simulations are $U_{\text{lid}} = 0.1$, $L = 1$, and $\rho = 1$. Once the Reynolds number is decided, kinematic viscosity υ can be calculated and the single relaxation parameter is then obtained from Eq. (19). Three cases are presented, with Re = 100, 400, and 1000, on a non-uniform mesh following the distribution rule of

$$x_i = 0.5\left[1 - \xi \tan^{-1}\left((1 - \kappa_i)\tan\left(1/\xi\right)\right)\right], \tag{90}$$

$$\kappa_i = \frac{i - 1}{2\left(i_{\max} - 1\right)}, \tag{91}$$

where i and $i_{\max}$ are the index of the mesh point and the total number of the mesh points along one direction, respectively. ξ is the mesh-stretching parameter, which is set to be 0.85 in this work. Note that due to the tie-up between the mesh spacing and time interval, extra interpolation algorithms are required. Specifically, the values of the equilibrium/non-equilibrium distribution functions on the positions where the streaming process initiates are evaluated from known values on neighboring mesh points where physical quantities are assigned. In our work, the classic Lagrange interpolation algorithm is applied, and $3 \times 3 \times 3$ neighboring mesh points are involved in the interpolation process to ensure the second-order spatial accuracy along any direction.

First, a convergence study is conducted on the case of Re = 1000. Various mesh sizes of $11 \times 11 \times 11$, $41 \times 41 \times 41$, $81 \times 81 \times 81$, and $121 \times 121 \times 121$ are applied. Velocity distribution along the centerlines is presented in Figure 5. It can be seen that curves obtained by mesh sizes of $81 \times 81 \times 81$ and $121 \times 121 \times 121$ generally overlap, which indicates that the computation converges well at mesh sizes finer than $81 \times 81 \times 81$. Therefore, in the following simulations, this mesh resolution is adopted. Another characteristic of the method reflected in the convergence test is its numerical stability. It is observed that at very coarse mesh resolution, e.g., $11 \times 11 \times 11$, the computation can still meet convergence criteria and obtain a smooth velocity field even if the results are not accurate enough. Comparatively, it has been reported in previous studies[27] that, at this Reynolds number, the conventional LBM exhibits unphysical oscillations when the mesh resolution is less than $61 \times 61 \times 61$. This indicates that the present method performs better than the conventional LBM in numerical stability, which also validates the theoretical analysis of the unconditional stability drawn in Section III C.

To numerically validate the advantage of 3D-USLBM in saving virtual memory, the memory cost of the 3D-USLBM

TABLE I. Comparisons of memory cost (unit: Megabytes) of the standard 3D-LBM and 3D-USLBM.

Methods	Mesh size		
	$41 \times 41 \times 41$	$81 \times 81 \times 81$	$121 \times 121 \times 121$
3D-LBM	28.8	219.4	730.3
3D-USLBM	15.2	114.0	378.9
3D-USLBM/3D-LBM	0.528	0.520	0.519

TABLE II. Comparison of the computational time of 3D-USLBM and 3D-LBM with converged results.

Methods	Mesh size	Computational time (s)	3D-USLBM/3D-LBM
3D-USLBM	$41 \times 31 \times 41$ (nonuniform)	3 833.1	0.699
	$81 \times 81 \times 81$ (uniform)	10 696.1	1.950
3D-LBM	$81 \times 81 \times 81$ (uniform)	5 485.9	...

and standard three-dimensional LBM (3D-LBM) at different mesh sizes is tabulated in Table I. It is clearly shown in this table that the memory cost of 3D-USLBM is roughly around 50% of the memory cost in 3D-LBM. The significant reduction in memory cost comes from the evolution characteristic of 3D-USLBM. The macroscopic variables, rather than the distribution functions, are directly updated and stored in the 3D-USLBM. Therefore, it is no longer necessary to store distribution functions on each mesh point, which saves a lot of virtual memory.

In Table II, we present the comparison of the computational cost of the 3D-LBM and 3D-USLBM. The Reynolds number is set to be Re = 400 here. The mesh sizes are chosen to ensure that the results are well converged. As can be seen, on a uniform mesh with the same mesh size, 3D-USLBM consumes more computational time than 3D-LBM, but the 3D-USLBM

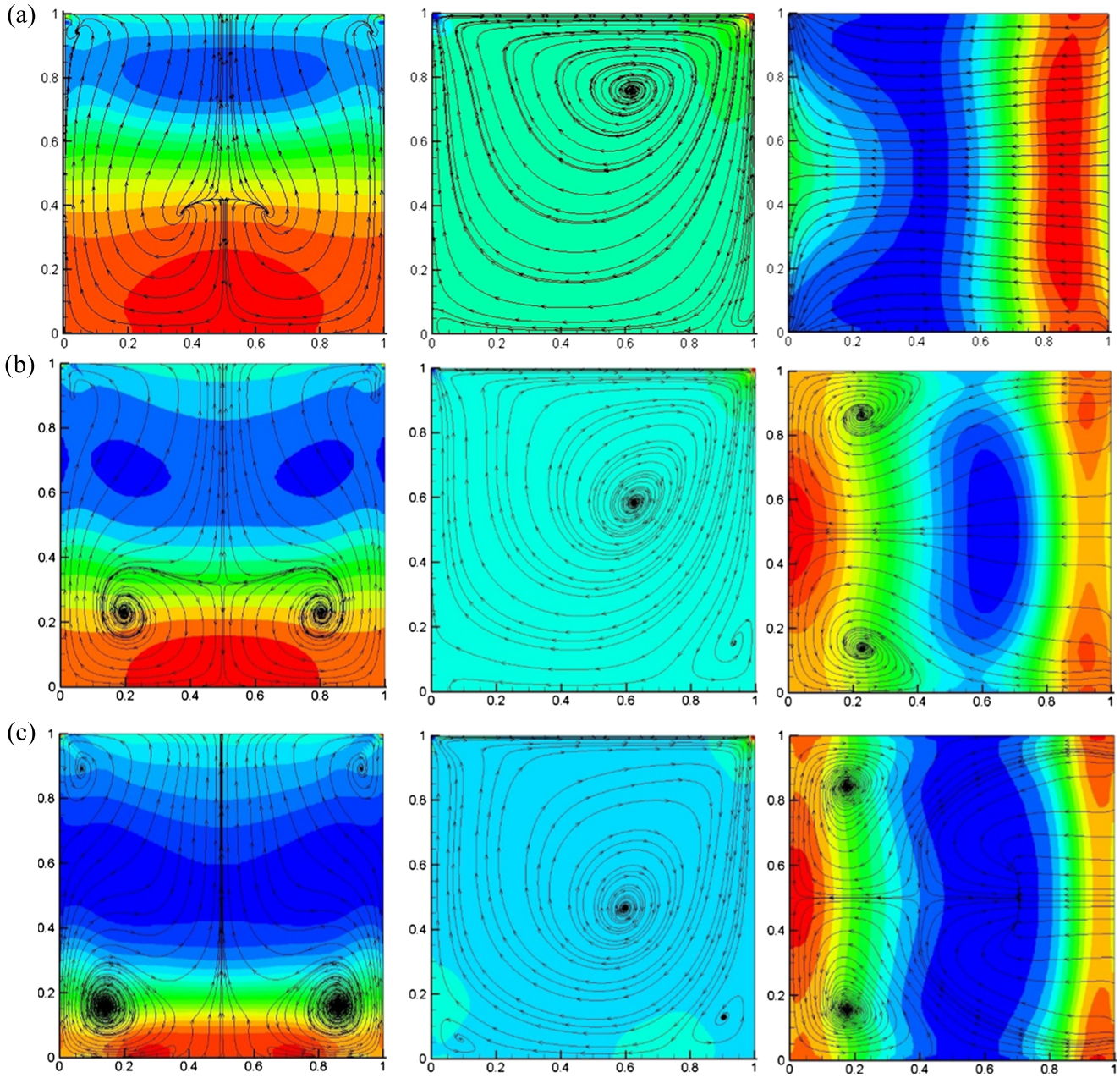


FIG. 6. Pressure contours and streamlines on the mid-planes. Left: x = 0.5; middle: y = 0.5; right: z = 0.5. (a) Re = 100; (b) Re = 400; (c) Re = 1000.

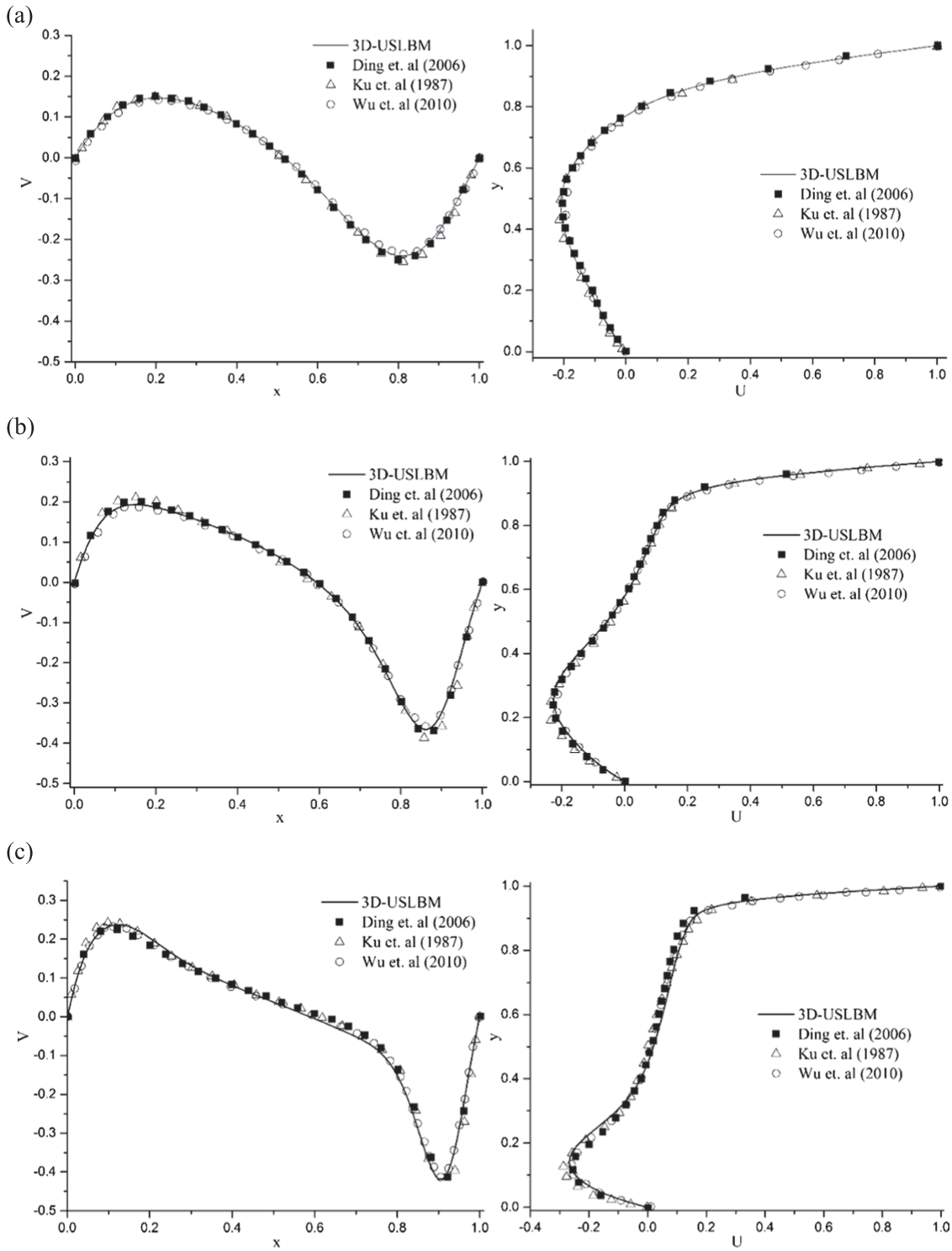


FIG. 7. Comparisons of the velocity distribution. Left: x-V curve; right: U-y curve. (a) Re = 100; (b) Re = 400; (c) Re = 1000.

can be flexibly extended to a non-uniform mesh. Therefore, fewer mesh points are required by 3D-USLBM to obtain converged results on non-uniform meshes. To obtain comparable results at the same Reynolds number, 3D-USLBM only needs a non-uniform mesh size of $41 \times 31 \times 41$. In such condition, the computational time of 3D-USLBM is only about 70% that of 3D-LBM. However, it should be noted that the computational time and the memory cost are the outcomes of compromises in application, which is largely dependent on tricks adopted in programming. Therefore, the comparisons conducted here are only of some reference values to the readers.

Figure 6 gives the pressure contours and the streamlines in the mid-planes at Re = 100, 400, and 1000. The pressure contours and the shape of streamlines obtained by 3D-USLBM agree well with those reported in previous studies.[27,52] Flow details like the main vortex and the small vortex at the corners at a higher Reynolds number are captured well by the present

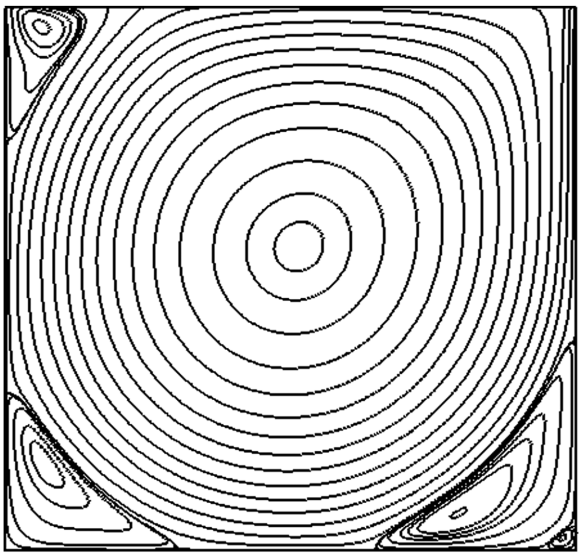


FIG. 8. Streamlines of the 2D lid-driven cavity flow at Re = 7500.

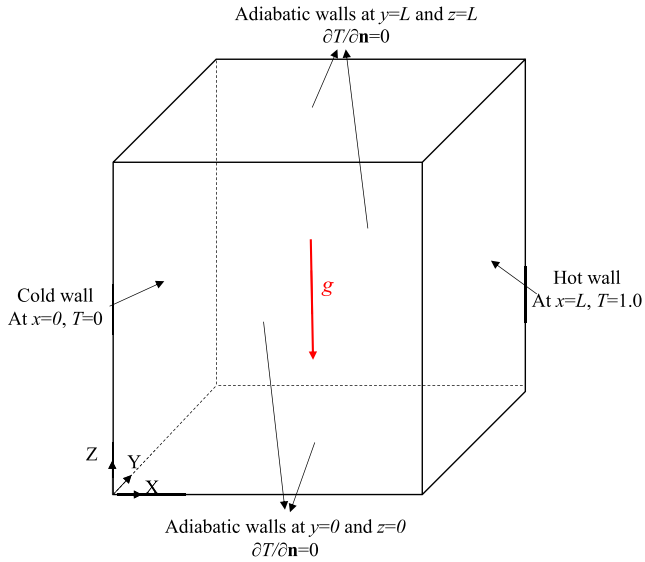


FIG. 9. Configuration of the 3D natural convection in a cubic cavity.

method. To quantitatively validate the results, the velocity profiles along the centerlines in our simulations are compared with those obtained by Ding *et al.*,[53] Ku *et al.*,[54] and Wu and Shu,[52] as presented in Figure 7. It is observed that the present results are in good agreement with these data. Therefore, it can be concluded that the present method gives reliable results in 3D simulations of incompressible isothermal viscous flows.

To show the ability of USLBM in capturing flow details more explicitly, here we also offer the numerical results in a 2D lid-driven cavity flow at Re = 7500. Under this condition, the velocity gradient is strong, and a very tiny vortex manifests at the bottom right corner. As presented in Figure 8, the present unconditionally stable lattice Boltzmann method well captures such flow details, which demonstrates its numerical accuracy.

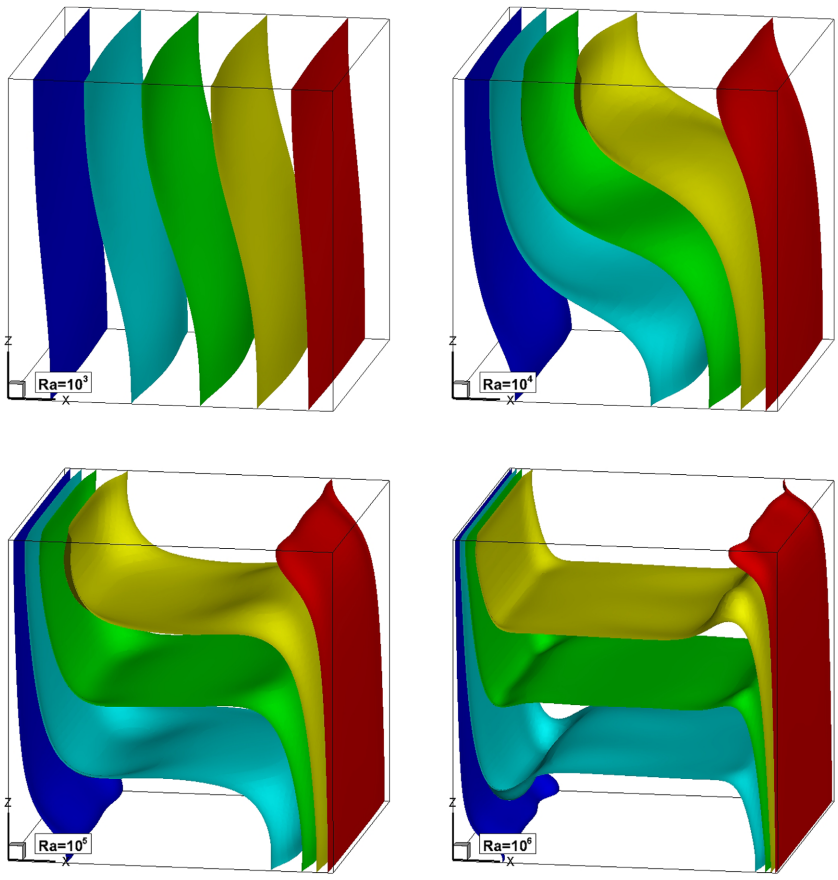


FIG. 10. Isotherm surfaces of the 3D natural convection in a cubic cavity at different Rayleigh numbers of 10^3, 10^4, 10^5, and 10^6.

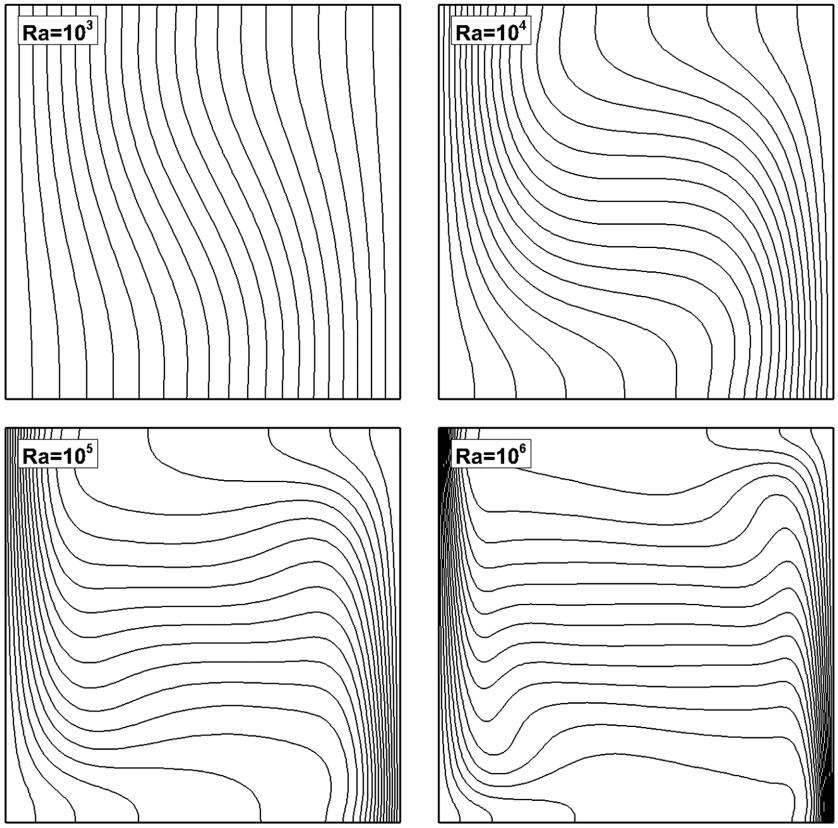


FIG. 11. Temperature contours in the symmetric plane of $y = 0.5$ for the 3D natural convection in a cubic cavity.

C. Natural convection in a cubic cavity

To show the adaptability of 3D-USLBM in simulating 3D thermal flows, the natural convection in a cubic cavity is studied in this section. As shown in Figure 9, the flow is driven by a temperature difference between the left and right walls. A natural boundary condition of temperature is implemented on the other walls. The final flow shape of this problem is determined by the Rayleigh number, defined as

$$\mathrm{Ra} = \frac{g\beta \cdot \Delta T \cdot L^3}{\upsilon \cdot \chi} = \frac{V_c^2 \cdot L^2}{\upsilon \cdot \chi}, \tag{92}$$

where L is the characteristic length of the square cavity, ΔT denotes the temperature difference between the hot and the cold walls, and $V_c = \sqrt{g\beta L \cdot \Delta T}$ is the characteristic thermal velocity constrained by the low Mach number limit. In this paper, we set $\mathrm{Pr} = \upsilon/\chi = 0.71$ and $V_c = 0.1$. Therefore, the

TABLE III. Comparison of flow properties in the symmetric plane at $y = 0.5$ of the 3D natural convection.

Ra		10^3	10^4	10^5	10^6
$u_{\max}$	3D-USLBM	0.132	0.198	0.140	0.077
	Peng *et al.*[48]	0.132	0.206	0.149	...
	Fusegi *et al.*[55]	0.131	0.201	0.147	0.084
	Wang *et al.*[27]	0.132	0.200	0.142	0.082
z	3D-USLBM	0.188	0.175	0.150	0.138
	Peng *et al.*[48]	0.188	0.163	0.136	...
	Fusegi *et al.*[55]	0.200	0.183	0.145	0.144
	Wang *et al.*[27]	0.187	0.176	0.147	0.146
$w_{\max}$	3D-USLBM	0.133	0.221	0.245	0.257
	Peng *et al.*[48]	0.133	0.221	0.240	...
	Fusegi *et al.*[55]	0.132	0.225	0.247	0.259
	Wang *et al.*[27]	0.133	0.221	0.244	0.253
x	3D-USLBM	0.825	0.883	0.933	0.964
	Peng *et al.*[48]	0.826	0.887	0.935	...
	Fusegi *et al.*[55]	0.833	0.883	0.935	0.967
	Wang *et al.*[27]	0.829	0.885	0.932	0.968
$\mathrm{Nu}_{\mathrm{mean}}$	3D-USLBM	1.087	2.250	4.623	8.801
	Peng *et al.*[48]	1.097	2.304	4.658	...
	Fusegi *et al.*[55]	1.105	2.302	4.646	9.012
	Wang *et al.*[27]	1.092	2.289	4.622	8.921
$\mathrm{Nu}_{\mathrm{overall}}$	3D-USLBM	1.069	2.054	4.361	8.601
	Ha and Jung[56]	1.072	2.07	4.464	...
	Fusegi *et al.*[55]	1.085	2.085	4.361	8.770
	Wang *et al.*[27]	1.071	2.062	4.344	8.684

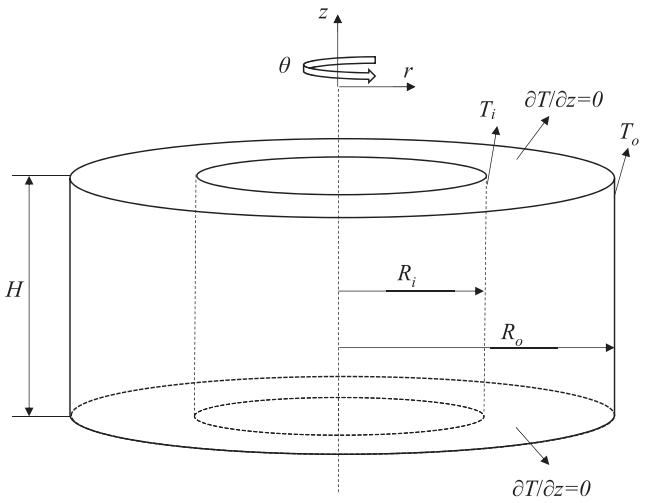


FIG. 12. Schematic diagram of the 3D natural convection in a concentric annulus.

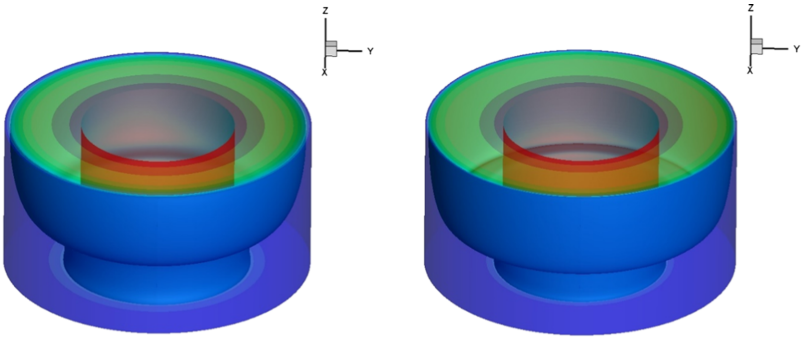


FIG. 13. Isotherm surfaces in the 3D natural convection in a concentric annulus.

kinematic viscosity υ and the diffusivity χ are determined once the specific value of Ra is given, and the single relaxation parameters τ_υ and τ_c are then obtained from Eqs. (19) and (20). To quantify the heat transfer rate, various Nusselt numbers have been defined. In this example, the mean Nusselt number and the overall Nusselt number on the heated wall are chosen for comparative studies, which are respectively defined as

$$Nu_{mean}(y) = \int_0^1 \left.\frac{\partial T(y,z)}{\partial x}\right|_{x=1} \mathrm{d}z, \tag{93}$$

$$Nu_{overall} = \int_0^1 Nu_{mean}(y)\,\mathrm{d}y. \tag{94}$$

Four cases with a wide range of Rayleigh numbers of 10^3, 10^4, 10^5, and 10^6 are simulated on a non-uniform mesh with a mesh size of $81 \times 41 \times 81$. Note that the distribution of the non-uniform mesh points also follows Eq. (90). As shown in Figure 10, smooth isotherms are obtained in all cases, which implies the good numerical stability of the present method. In Figure 11, the isotherms on the symmetric plane of $y = 0.5$ are depicted. It is observed that the isotherms are squeezed more to the left/right boundaries at higher Rayleigh numbers, which indicates a larger heat transfer rate in these areas. For the purpose of quantitative comparison, the representative hydrodynamic parameters and the average Nusselt number over the whole domain are tabulated in Table III. Parameters considered here include the maximum horizontal velocity $u_{\max}$ along the vertical centerline and its position, and the maximum vertical velocity $w_{\max}$ on the horizontal centerline and its location. Consistent with observations from the isotherm pictures, the Nusselt number does indeed increase with the Rayleigh number. Comparisons with the data reported in the literature[27,48,55] are made, for which the maximum relative error is found to be within 1.5%. Such performance demonstrates the high accuracy of the present method. Together with its flexibility on a non-uniform mesh as shown in this example, 3D-USLBM is a very promising method in simulating three-dimensional incompressible thermal flows.

D. Natural convection in a concentric annulus

In this section, the example of the 3D natural convection in a concentric annulus is tested to evaluate the present 3D-USLBM in problems with body-fitted meshes. The physical configuration of this problem is presented in Figure 12. Constant temperatures T_i and T_o are imposed on the inner and outer cylinder surfaces, respectively, while an adiabatic condition with $\partial T/\partial z = 0$ is implemented on both the top and the bottom boundaries. Radii of the inner and the outer cylinder are denoted as R_i and R_o, respectively, and H represents the height of the annulus. The flow shape of this problem is

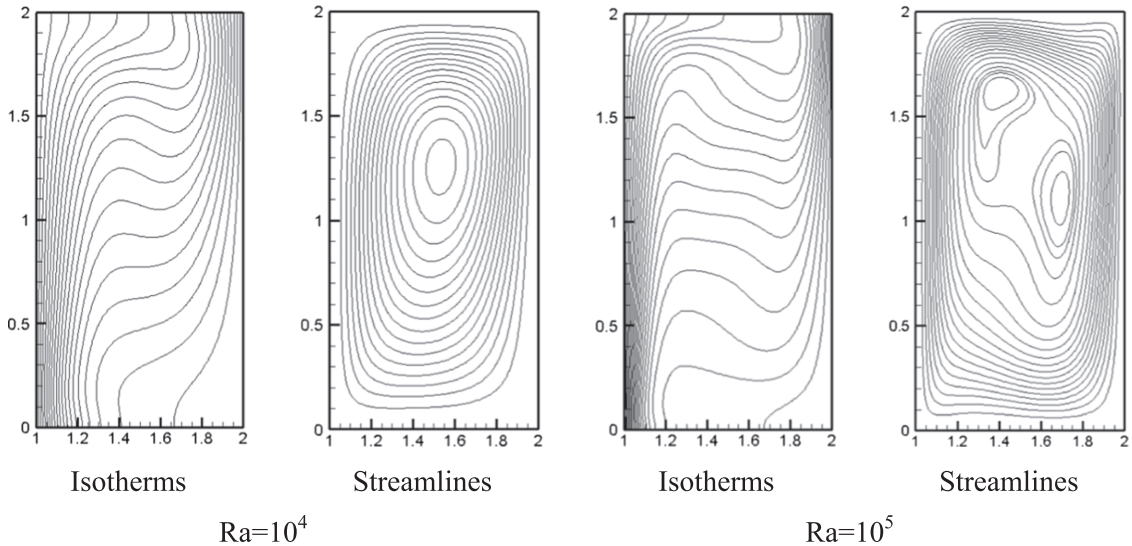


FIG. 14. Isotherms and streamlines of the r-z plane for the 3D natural convection in a concentric annulus.

TABLE IV. Comparison of average Nusselt numbers for the 3D natural convection in a concentric annulus.

Ra	Venkatachalappa *et al.*[59] $\left(\overline{\mathrm{Nu}_i}+\overline{\mathrm{Nu}_o}\right)/2$	Li *et al.*[60] $\overline{\mathrm{Nu}_i}$	Li *et al.*[60] $\overline{\mathrm{Nu}_o}$	Wang *et al.*[61] $\overline{\mathrm{Nu}_i}$	Wang *et al.*[61] $\overline{\mathrm{Nu}_o}$	Present $\overline{\mathrm{Nu}_i}$	Present $\overline{\mathrm{Nu}_o}$
10^4	3.163	3.126	3.216	3.220	3.217	3.228	3.193
10^5	5.882	5.798	5.798	5.815	5.808	5.731	5.712

determined by four non-dimensional parameters which include two aspect ratios $\mathrm{Ar}_{io} = R_o/R_i$ and $\eta = H/(R_o - R_i)$, Prandtl number $\mathrm{Pr} = \upsilon/\chi$, and Rayleigh number $\mathrm{Ra} = g\beta(T_i - T_o)\,(R_o - R_i)^3/(\upsilon\chi)$. In the present study, we set $\mathrm{Ar}_{io} = \eta = 2$ and $\mathrm{Pr} = 0.7$. The characteristic velocity $\sqrt{g\beta\,(T_i - T_o)\,(R_o - R_i)}$ is chosen as 0.1 to meet the incompressible limit. Similar to previous examples, single relaxation parameters are decided from kinematic viscosity υ and diffusivity χ which are determined by specified values of Ra. Two cases with Rayleigh numbers of 10^4 and 10^5 are considered, and a mesh size of $101 \times 101 \times 201$ is adopted. Note that although the mesh spacing is uniform along a given direction, interpolation is still inevitable because the mesh is curved along the azimuthal direction and the initial positions of the streaming process may not coincide with the mesh points. Specifically, in our simulations, the coordinates of the initial positions for streaming in the computational domain can be calculated once the streaming distance is determined. Consequently, the Lagrangian interpolation algorithm is implemented along the radial, the azimuthal, and the vertical directions to evaluate functional values at those positions.

Isotherm surfaces at the converged state are depicted in Figure 13. It is shown in this figure that the resultant contour is axisymmetric about the central axis of the concentric annulus, which is in accordance with the common knowledge on this problem.[57,58] Isotherms and streamlines of the r-z plane are presented in Figure 14 to give a detailed picture of the temperature distribution and flow pattern. As can be seen, smooth temperature fields and streamline patterns are obtained, which demonstrates the stability of the combined solver of 3D-USLBM and Lagrangian interpolation. It is noted that, at a higher Rayleigh number ($\mathrm{Ra} = 10^5$), isotherms are squeezed more towards the hot/cold walls. Such performance represents the larger temperature gradient and higher heat transfer rate. We also notice that a pair of small vortices is shown in the case at a higher Rayleigh number, which indicates that more severe flow patterns can be expected at higher Rayleigh numbers.

To quantify the heat transfer rate in this problem, the following average Nusselt numbers in the inner and the outer cylinders are defined as follows:

$$\overline{Nu}_{i,o} = -\frac{R_{i,o}}{H}\int_0^H \left.\frac{\partial T}{\partial r}\right|_{i,o} \mathrm{d}z. \tag{95}$$

In Table IV, the calculated Nusselt numbers in our simulations are tabulated. The average Nusselt number is larger at a higher Rayleigh number, which represents a higher heat transfer rate. Such results are in accordance with our observations of the isotherm contours. Comparing with data given by Venkatachalappa *et al.*,[59] Li *et al.*,[60] and Wang *et al.*,[61] it turns out that the maximum relative error of our results is within 1.5%, which validates the robustness and flexibility of 3D-USLBM in thermal problems with a body-fitted non-uniform mesh.

V. CONCLUDING REMARKS

The standard lattice Boltzmann method (LBM) is a popular numerical method. However, severe drawbacks, e.g., high cost in virtual memory, inconvenient implementation of physical boundary conditions, and poor numerical stability, are witnessed in the standard LBM. To alleviate these issues, a simplified lattice Boltzmann method (SLBM) was recently proposed. However, SLBM can only manage to achieve the first-order of accuracy in time. Moreover, its robustness in three-dimensional scenarios has not been validated yet.

In this paper, a three-dimensional simplified and unconditionally stable lattice Boltzmann method (3D-USLBM) with the second-order of accuracy both in time and in space is developed for the simulation of three-dimensional incompressible isothermal/thermal flows. The macroscopic governing equations recovered from the lattice Boltzmann equation (LBE) through the Chapman-Enskog (C-E) expansion analysis can be resolved in a predictor-corrector step by employing the fractional step technique. The solutions to these equations are reconstructed within the LBM framework by using lattice properties and relationships given by the C-E analysis to obtain final formulations of 3D-USLBM. In 3D-USLBM, the equilibrium distribution functions are calculated from the macroscopic variables, while the non-equilibrium distribution functions are evaluated from the difference of two equilibrium distribution functions at different locations and time levels. Therefore, in 3D-USLBM, the macroscopic variables rather than the distribution functions are tracked in the computational process, which greatly saves the virtual memory and facilitates the implementation of physical boundary conditions.

As its name says, the presented method has good stability performance. Through the von Neumann stability analysis, 3D-USLBM can be proven to be unconditionally stable. Such a characteristic makes it a promising replacement for the standard lattice Boltzmann method in solving high Reynolds/Rayleigh number flow problems. This theoretical conclusion is further supported by stability tests on the lid-driven cavity flow. It turns out that even for very coarse meshes, this method can still trigger the convergence criteria.

Four representative three-dimensional problems, including plane Poiseuille flow, lid-driven cavity flow, natural convection in a cubic cavity, and natural convection in a concentric

annulus, are simulated to provide comprehensive validations of the accuracy, convergence, and flexibility of the proposed method. Good agreements with reference data are observed in these tests.

In summary, the presented three-dimensional simplified and unconditionally stable lattice Boltzmann method (3D-USLBM) is a truly second-order scheme within the LBM framework. Compared to the standard LBM with the second-order of accuracy, 3D-USLBM greatly saves virtual memory and facilitates the implementation of physical boundary conditions. Its unconditionally stable characteristic *makes it highly suitable for* applications in which high Reynolds/Rayleigh numbers are usually encountered.

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